Geometry & Measurement

Angle facts

1. Two angles on a straight line measure $3x$ and $2x + 30$. Find x.

2. Find the angle vertically opposite a 65° angle.

3. Three angles around a point measure 110°, 95°, and x. Find x.

4. Two angles are complementary (sum to 90°). One measures 37°. Find the other.

Parallel lines and transversals

5. Two parallel lines are cut by a transversal, creating an angle of 70°. Find its corresponding angle.

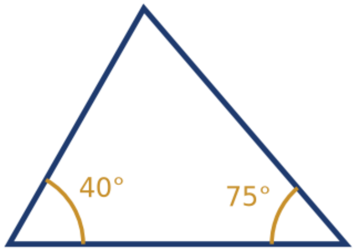
6. In the same setup, find the co-interior (allied) angle to a 70° angle.

7. Find the alternate interior angle to a 55° angle.

8. Two parallel lines cut by a transversal create an angle of $4x$ and its co-interior angle of $2x + 30$. Find x.

Triangle angle sum

9. Find the third angle of the triangle shown, given the two marked angles.



10. An isosceles triangle has two equal angles of 72° each. Find the third angle.

11. In a right triangle, one of the non-right angles is 35°. Find the other non-right angle.

12. A triangle has three equal angles. Find each angle, and name this type of triangle.

Angle sums of polygons

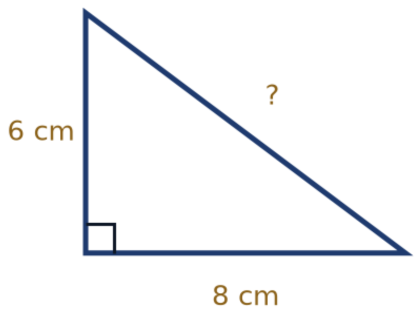
13 Find the sum of the interior angles of a hexagon (6 sides).

14 Find the sum of the interior angles of a pentagon (5 sides).

15 A regular polygon has an interior angle sum of $1{,}080^\circ$. How many sides does it have?

The Pythagorean theorem

16 Find the hypotenuse of the right triangle shown.



17 A right triangle has hypotenuse 13 and one leg 5. Find the other leg.

18 A 10 m ladder leans against a wall with its foot 6 m from the wall's base. How high up the wall does it reach?

19 A right triangle has legs 9 and 12. Find the hypotenuse.

Perimeter and area of rectangles and squares

20 A rectangle has length 12 cm and width 7 cm. Find its perimeter and area.

21 A square has side length 9 cm. Find its perimeter and area.

22 A rectangle has perimeter 50 cm and length 16 cm. Find its width and area.

23 A square has area $121\ \text{cm}^2$. Find its side length and perimeter.

Area of triangles and parallelograms

24 A triangle has base 14 cm and height 9 cm. Find its area.

25 A parallelogram has base 12 cm and height 8 cm. Find its area.

26 A triangle has area 60 cm^2 and base 15 cm. Find its height.

27 A parallelogram has area 126 cm^2 and height 9 cm. Find its base.

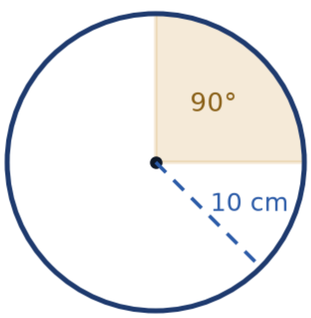
Circles: circumference and area

28 A circle has radius 7 cm. Find its circumference, using $\pi = \frac{22}{7}$.

29 A circle has radius 7 cm. Find its area, using $\pi = \frac{22}{7}$.

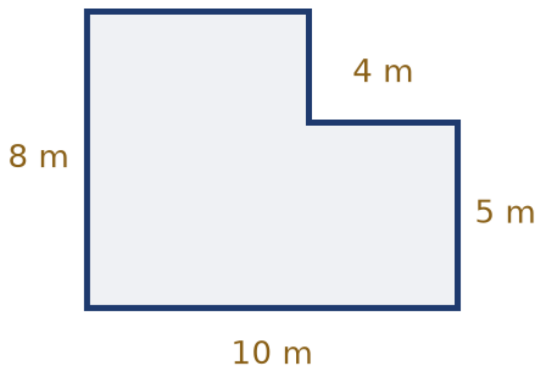
30 A circle has diameter 20 cm. Find its circumference in terms of π.

31 Find the area of the shaded 90° sector shown, in terms of π.



Composite figures

32 Find the area of the L-shaped room shown.



33 A garden is L-shaped: a 15 m by 6 m rectangle with a 5 m by 2 m rectangle cut from one corner. Find its area.

34 A shape consists of a 12 cm by 5 cm rectangle with a right triangle of base 6 cm and height 5 cm attached to one of the short sides. Find the total area.

Volume and surface area

35 A rectangular box has dimensions 5 cm $\times 4$ cm $\times 3$ cm. Find its volume.

36 A cube has edge length 6 cm. Find its volume and surface area.

37 A rectangular prism has volume 240 cm^3, length 10 cm, and width 6 cm. Find its height.

38 Find the surface area of a rectangular box with dimensions 8 cm $\times 5$ cm $\times 4$ cm.

Word problems

39 A rectangular garden measuring 20 m by 15 m is surrounded by a path 1 m wide. Find the area of the path alone.

40 New flooring costs 45 riyals per square meter. Find the total cost to floor a rectangular room measuring 6 m by 4 m.

41 A 17 m ladder leans against a vertical wall with its base 8 m from the wall. How high up the wall does it reach?

42 A jogger runs 4 laps of a circular track of radius 35 m, using $\pi = \frac{22}{7}$. Find the total distance covered.

MCQ — Angle facts

43 Two angles are supplementary. One is three times the other. Find the smaller angle.

a. 60°

b. 30°

c. 135°

d. 45°

44 Three angles around a point measure 118°, 142°, and x. Find x.

a. 80°

b. 140°

c. 120°

d. 100°

45 An angle is four times its complement. Find the angle.

a. 54°

b. 18°

c. 72°

d. 90°

46 Two angles on a straight line measure $(4x + 10)^\circ$ and $(2x + 20)^\circ$. Find x.

a. 20

b. 35

c. 15

d. 25

MCQ — Parallel lines and transversals

47 Two parallel lines are cut by a transversal. One angle is $(3x+15)^\circ$ and its corresponding angle is $(5x-25)^\circ$. Find x.

a. 10

b. 8

c. 30

d. 20

48 Two parallel lines cut by a transversal create co-interior angles of $(2x+10)^\circ$ and $(3x-30)^\circ$. Find x.

a. 30

b. 40

c. 50

d. 34

49 Two parallel lines are cut by a transversal. One angle is 63°. Find its alternate exterior angle.

a. 153°

b. 27°

c. 63°

d. 117°

50 Two parallel lines are cut by a transversal. One angle and its co-interior angle differ by 40° and sum to 180°. Find the larger angle.

a. 110°

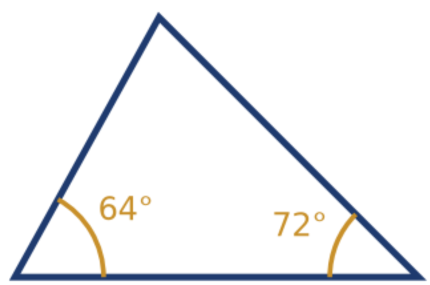
b. 70°

c. 140°

d. 90°

MCQ — Triangle angle sum

51 Find the third angle of the triangle shown.



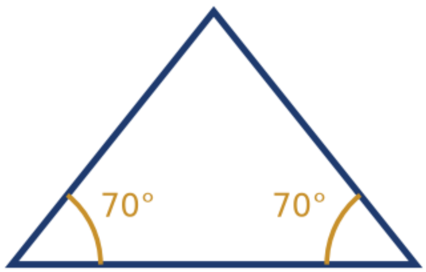
a. 116°

b. 64°

c. 54°

d. 44°

52 The isosceles triangle shown has two equal base angles of 70° each. Find the apex angle.



a. 55°

b. 70°

c. 40°

d. 110°

53 A triangle has three angles in the ratio $2:3:4$. Find the largest angle.

a. 90°

b. 100°

c. 80°

d. 60°

54 In a right triangle, one non-right angle exceeds the other by 26°. Find the larger non-right angle.

a. 32°

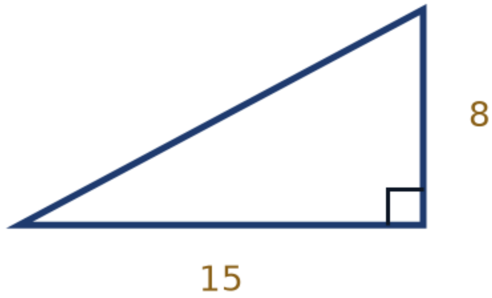
b. 58°

c. 64°

d. 48°

MCQ — The Pythagorean theorem

55 Find the hypotenuse of the right triangle shown.



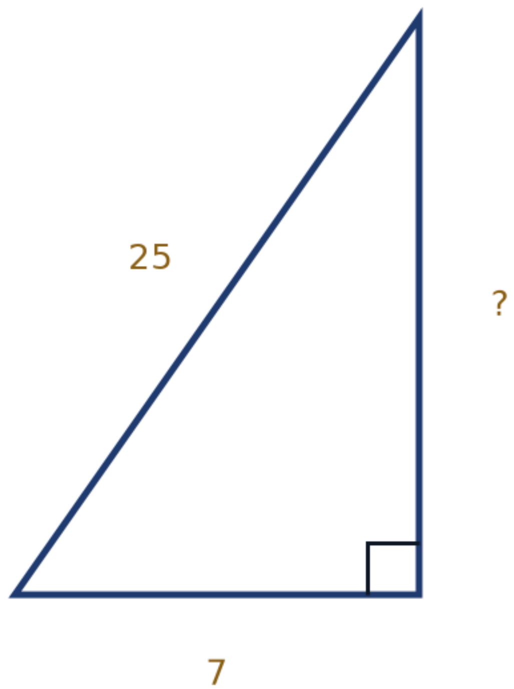
a. 23

b. 19

c. 17

d. 13

56 Find the missing leg of the right triangle shown.



a. 26

b. 18

c. 24

d. 20

57 A rectangular field is 24 m long and 18 m wide. Find the length of its diagonal.

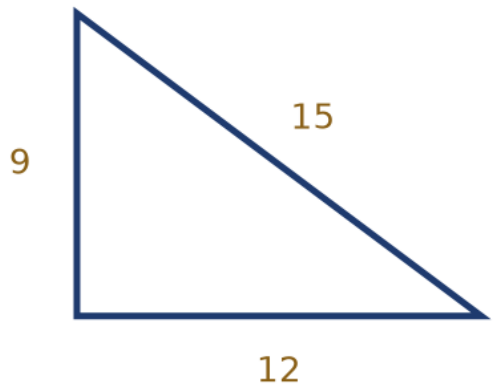
a. 30

b. 36

c. 42

d. 26

58 A triangle has sides 9 cm, 12 cm, and 15 cm, as shown. Is it a right triangle?



a. Only if it is also isosceles

b. Yes — a right triangle

c. Cannot be determined

d. No — not a right triangle

59 A 15 m ladder leans against a wall with its foot 9 m from the wall's base. How high up the wall does the ladder reach?

a. 13

b. 10

c. 12

d. 18

MCQ — Perimeter and area of rectangles and squares

60 Find the perimeter and area of the rectangle shown.

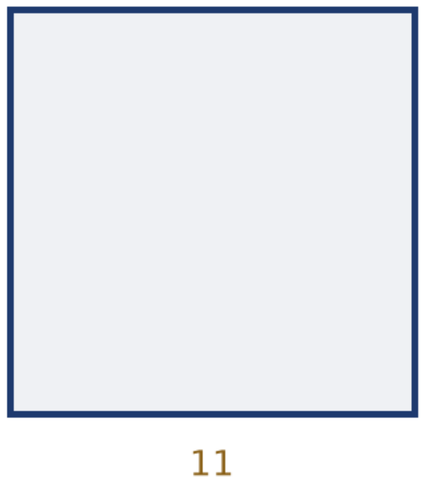
a. $P = 48$ cm, $A = 135$ cm^2

b. $P = 48$ cm, $A = 24$ cm^2

c. $P = 135$ cm, $A = 48$ cm^2

d. $P = 24$ cm, $A = 135$ cm^2

61 Find the perimeter and area of the square shown.



a. $P = 44$ cm, $A = 110$ cm^2

b. $P = 44$ cm, $A = 121$ cm^2

c. $P = 22$ cm, $A = 121$ cm^2

d. $P = 121$ cm, $A = 44$ cm^2

62 A rectangle has perimeter 64 cm and length 20 cm. Find its width and area.

a. $w = 12$ cm, $A = 32$ cm^2

b. $w = 12$ cm, $A = 240$ cm^2

c. $w = 44$ cm, $A = 240$ cm^2

d. $w = 22$ cm, $A = 440$ cm^2

63 A square has area 225 cm^2. Find its side length and perimeter.

a. $s = 15$ cm, $P = 30$ cm

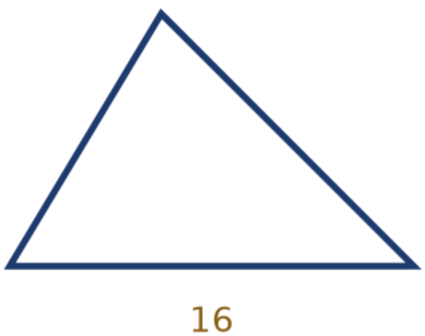
b. $s = 225$ cm, $P = 900$ cm

c. $s = 15$ cm, $P = 60$ cm

d. $s = 45$ cm, $P = 60$ cm

MCQ — Area of triangles and parallelograms

64 The triangle shown has base 16 cm and height 10 cm. Find its area.



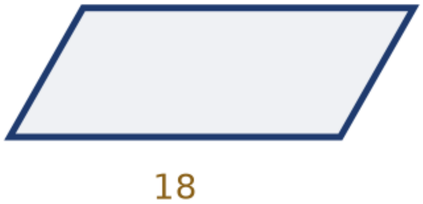
a. 160

b. 26

c. 80

d. 40

65 The parallelogram shown has base 18 cm and height 7 cm. Find its area.



a. 252

b. 126

c. 63

d. 25

66 A triangle has area 84 cm^2 and height 12 cm. Find its base.

a. 14

b. 1,008

c. 28

d. 7

67 A parallelogram has area 150 cm^2 and base 15 cm. Find its height.

a. 20

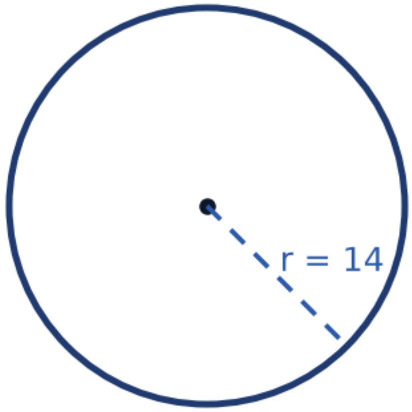
b. 10

c. 2,250

d. 5

MCQ — Circles: circumference and area

68 Find the circumference of the circle shown, using $\pi = \frac{22}{7}$.



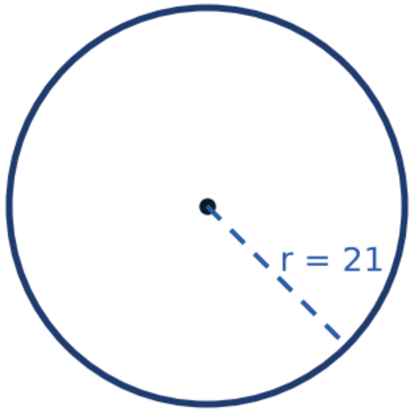
a. 44

b. 616

c. 88

d. 176

69 Find the area of the circle shown, using $\pi = \frac{22}{7}$.



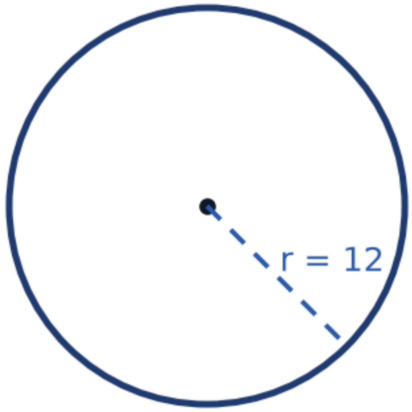
a. 132

b. 2,772

c. 693

d. 1,386

70 Find the circumference of the circle shown, in terms of π.



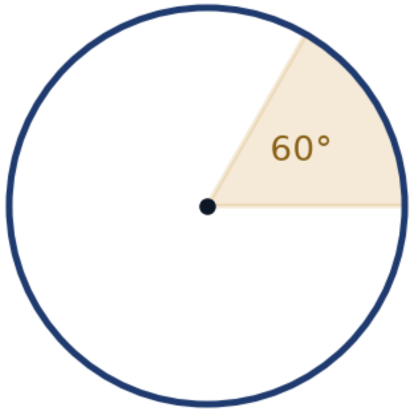
a. 48π cm

b. 24π cm

c. 12π cm

d. 144π cm

71 Find the area of the shaded 60° sector shown, in terms of π.



a. 81π cm^2

b. 27π cm^2

c. 13.5π cm^2

d. 4.5π cm^2

72 A circle has area 144π cm^2. Find its radius.

a. 6

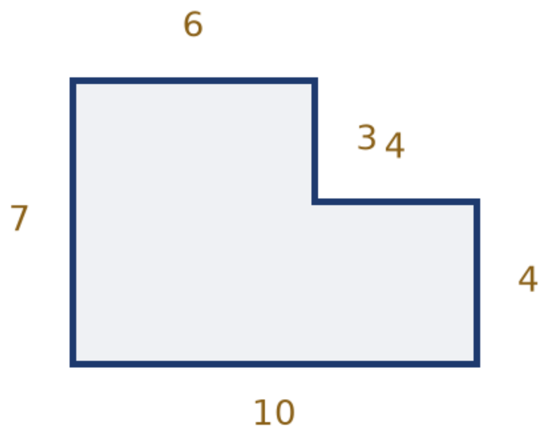
b. 24

c. 12

d. 72

MCQ — Composite figures

73 Find the area of the L-shaped room shown (all measurements in meters).



a. 70

b. 58

c. 46

d. 82

74 A garden is L-shaped: an 18 m by 8 m rectangle with a 6 m by 3 m rectangle removed from one corner. Find its area.

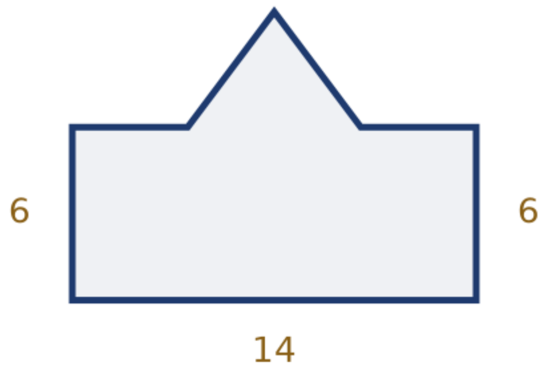
a. 108

b. 126

c. 162

d. 138

75 Find the area of the shape shown: a 14 cm by 6 cm rectangle with a right triangle of base 6 cm and height 4 cm attached to one side.



a. 108

b. 90

c. 96

d. 72

76 A rectangular lawn 20 m by 12 m has a circular flower bed of radius 3.5 m removed from it. Find the remaining lawn area, using $\pi = \frac{22}{7}$.

a. 196

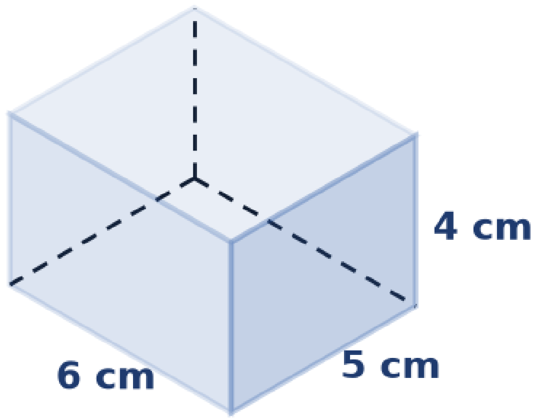
b. 278.5

c. 240

d. 201.5

MCQ — Volume and surface area

77 Find the volume of the rectangular box shown.



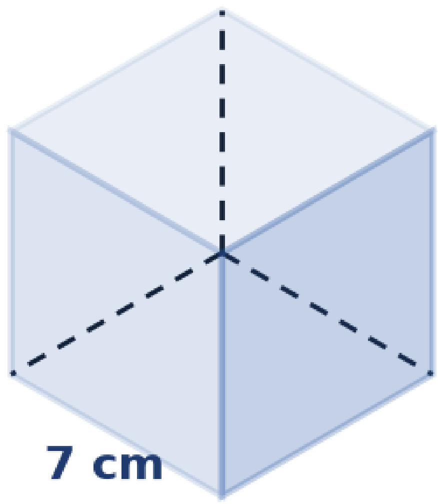
a. 120

b. 60

c. 15

d. 30

78 Find the volume and surface area of the cube shown.



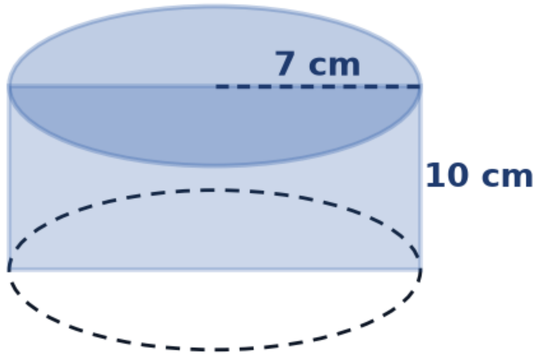
a. $V = 343\ \text{cm}^3$, $SA = 49\ \text{cm}^2$

b. $V = 49\ \text{cm}^3$, $SA = 294\ \text{cm}^2$

c. $V = 294\ \text{cm}^3$, $SA = 343\ \text{cm}^2$

d. $V = 343\ \text{cm}^3$, $SA = 294\ \text{cm}^2$

79 Find the volume of the cylinder shown, using $\pi = \frac{22}{7}$.



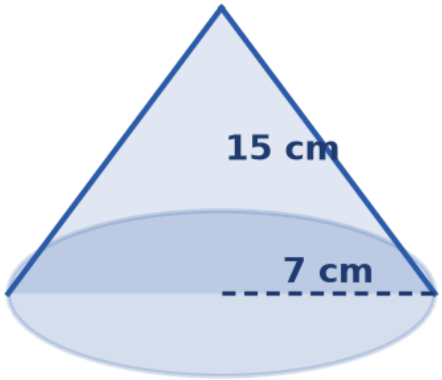
a. 1,540

b. 770

c. 440

d. 3,080

80 Find the volume of the cone shown, using $\pi = \frac{22}{7}$.



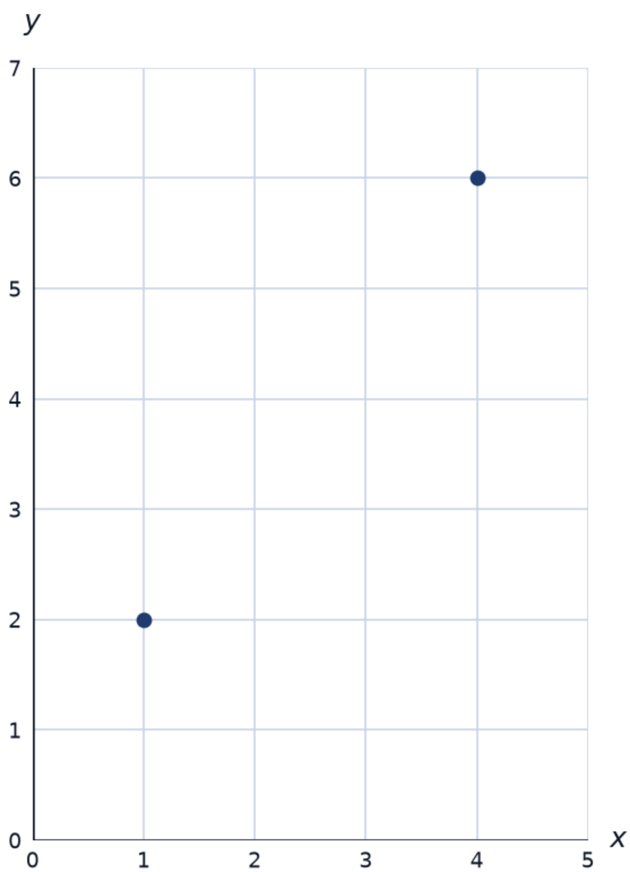
- **a.** 770
- **b.** 2,310
- **c.** 1,540
- **d.** 385

81 A rectangular prism has volume $360\ \text{cm}^3$, length 12 cm, and width 5 cm. Find its height.

- **a.** 6
- **b.** 15
- **c.** 72
- **d.** 30

MCQ — Coordinate geometry: distance and midpoint

82 Find the distance between the two points shown.



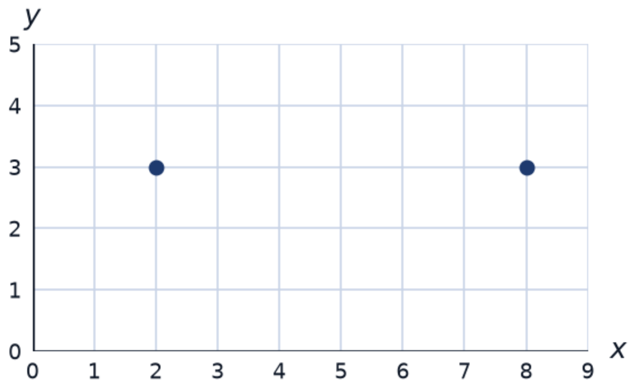
a. 5

b. 25

c. 7

d. $\sqrt{7}$

83 Find the midpoint of the segment joining the two points shown.



a. $(6, 3)$

b. $(3, 5)$

c. $(10, 6)$

d. $(5, 3)$

84 Find the distance between the points $(0, 0)$ and $(6, 8)$.

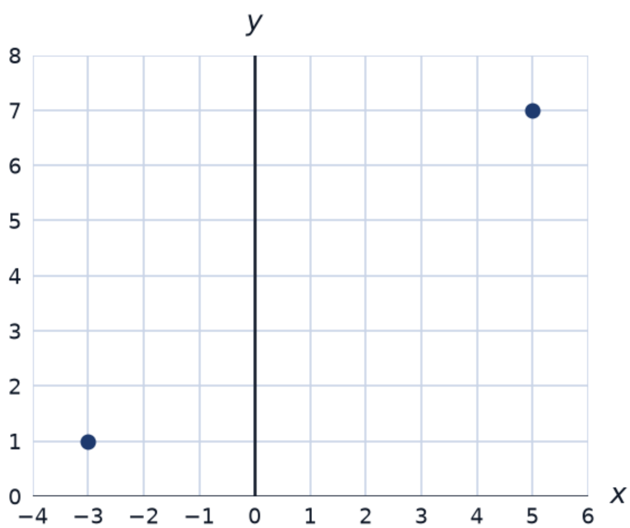
a. 14

b. 10

c. 7

d. 48

85 Find the midpoint of the segment joining the two points shown.



a. $(-1, -4)$

b. $(4, 1)$

c. $(2, 8)$

d. $(1, 4)$

MCQ — Similar figures and scale factor

86 Two similar triangles have a scale factor of $3 : 5$. If the smaller triangle's perimeter is 24 cm, find the larger triangle's perimeter.

a. 14.4

b. 72

c. 64

d. 40

87 A model car is built at a scale of $1:24$. If the model is 8 cm long, find the actual car's length in meters.

a. 19.2

b. 0.192

c. 1.92

d. 24

88 Two similar rectangles have areas in ratio $4:9$. Find the ratio of their side lengths.

a. $4:9$

b. $2:3$

c. $8:18$

d. $16:81$

89 A photo is enlarged by a scale factor of 2.5. If the original photo is 6 cm wide, find the width of the enlargement.

a. 8.5

b. 12.5

c. 3.6

d. 15

MCQ — Word problems

90 A rectangular pool measuring 25 m by 12 m is surrounded by a deck 2 m wide. Find the area of the deck alone.

a. 74

b. 300

c. 164

d. 464

91 Tiling costs 38 riyals per square meter. Find the total cost to tile a rectangular room measuring 9 m by 6 m.

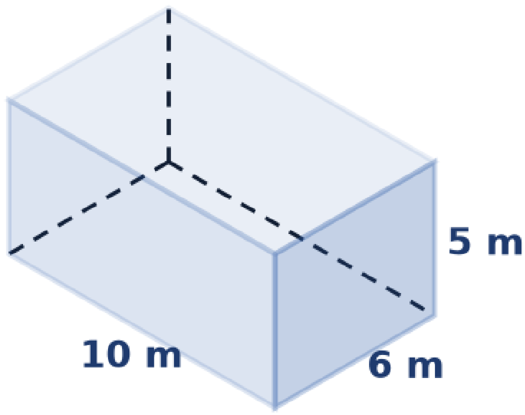
a. 1,026 SAR

b. 4,104 SAR

c. 342 SAR

d. 2,052 SAR

92 A water tank shaped like the rectangular box shown has these dimensions in meters. Find its capacity in cubic meters.



a. 150

b. 60

c. 300

d. 21